ADDITIONAL MATERIAL

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Lemma. *(See [CT26a, Lemma 6A.5]). Given the color basis, we have the following:*

- **(a)** $p_k \cdot p_m = \delta_{km} p_m$, *and hence* $E_{k,0} \cdot E_{m,0} = \delta_{km} E_{m,0}$ *and* $E_{k,\pm} \cdot E_{m,\pm} = \delta_{km} E_{m,\pm}$, *while all other products between color basis elements are zero;*
- **(b)** $\Delta_{n,1}(E_{k,0}) = -\frac{1}{3} f_{n,k} E_{k,0} \otimes E_{k,0}$ *and* $\Delta_{n,1}(E_{k,\pm}) = \frac{2}{3} f_{n,k} E_{k,\pm} \otimes E_{k,\pm}$;
- **(c)** $m_{n,1}(E_{k,0}) = u g_{n,k} E_{k,0}$ *and* $m_{n,1}(E_{k,\pm}) = (u+w) g_{n,k} E_{k,\pm}$,

where

$$f_{n,k} = \begin{cases} 1 & n \text{ even} \\ \omega^k & n \text{ odd} \end{cases}, \qquad g_{n,k} = \begin{cases} \omega^{kn/2} & n \text{ even} \\ \omega^{k(n+1)/2} & n \text{ odd} \end{cases}.$$

In other words, $\partial_{n,1}$ *is diagonal in the color basis.*

Proof. **(a)** The calcuation is straightfoward using the following result:

$$p_k \, x^{n-\ell} = \omega^{-k\ell} p_k$$

for all $\ell = 1, \dots, n$.

For **(b)**, let us use the fact that $\xi = 1$ to simplify the expression for $\Delta_{n,1}$:

$$\begin{aligned} \Delta_{n,1}(v) = &- \frac{1}{3n} \sum_{i+j \equiv c_n \bmod n} (v \cdot x^i) \otimes x^j \\ &+ \frac{1}{3n} \sum_{i+j \equiv (c_n - 1) \bmod n} (v \cdot x^i y^2) \otimes x^j + (v \cdot x^i y) \otimes x^j y + (v \cdot x^i) \otimes x^j y^2. \end{aligned}$$

for $c_n = \begin{cases} 0 & n \text{ even} \\ 1 & n \text{ odd} \end{cases}$.

We now show that $\Delta_{n,1}(E_{k,0}) = -\frac{1}{3} f_{n,k} E_{k,0} \otimes E_{k,0}$. The other cases involving $E_{k,\pm}$ proceed similarly.

Firstly,

$$\begin{aligned} E_{k,0} \otimes E_{k,0} &= p_k(1 - x^{n-1}y^2) \otimes p_k(1 - x^{n-1}y^2) \\ &= p_k \otimes p_k - \omega^{-k} p_k \otimes p_k y^2 - \omega^{-k} p_k y^2 \otimes p_k + \omega^{-2k} p_k y^2 \otimes p_k y^2 \\ &= \frac{1}{n^2} \sum_{r,s=0}^{n-1} \omega^{-k(r+s)} x^r \otimes x^s - \frac{1}{n^2} \sum_{r,s=0}^{n-1} \omega^{-k(r+s+1)} x^r \otimes x^s y^2 \\ &- \frac{1}{n^2} \sum_{r,s=0}^{n-1} \omega^{-k(r+s+1)} x^r y^2 \otimes x^s + \frac{1}{n^2} \sum_{r,s=0}^{n-1} \omega^{-k(r+s+2)} x^r y^2 \otimes x^s y^2 \\ &= \frac{1}{n^2} \sum_{\substack{\ell,r,s=0 \\ r+s \equiv \ell \bmod n}}^{n-1} \omega^{-k\ell} x^r \otimes x^s - \frac{1}{n^2} \sum_{\substack{\ell,r,s=0 \\ r+s+1 \equiv \ell \bmod n}}^{n-1} \omega^{-k\ell} (x^r \otimes x^s y^2 + x^r y^2 \otimes x^s) \\ &+ \frac{1}{n^2} \sum_{\substack{\ell,r,s=0 \\ r+s+2 \equiv \ell \bmod n}}^{n-1} \omega^{-k\ell} x^r y^2 \otimes x^s y^2. \end{aligned}$$

On the other hand,

$$\Delta_{n,1}(E_{k,0}) = \Delta_{n,1}(p_k) - \omega^{-k} \Delta_{n,1}(p_k y^2)$$

$$= -\frac{1}{3n^2} \sum_{\substack{\ell,i,j=0 \\ i+j\equiv c_n \bmod n}}^{n-1} \omega^{-k\ell} x^{\ell+i} \otimes x^j + \frac{1}{3n^2} \sum_{\substack{\ell,i,j=0 \\ i+j\equiv (c_n-1) \bmod n}}^{n-1} \omega^{-k\ell} x^{\ell+i} y^2 \otimes x^j$$

$$+ \frac{1}{3n^2} \sum_{\substack{\ell,i,j=0 \\ i+j\equiv (c_n-1) \bmod n}}^{n-1} \omega^{-k\ell} (x^{\ell+i} y \otimes x^j y + x^{\ell+i} \otimes x^j y^2)$$

$$+ \frac{1}{3n^2} \sum_{\substack{\ell,i,j=0 \\ i+j\equiv c_n \bmod n}}^{n-1} \omega^{-k(\ell+1)} x^{\ell+i} y^2 \otimes x^j - \frac{1}{3n^2} \sum_{\substack{\ell,i,j=0 \\ i+j\equiv (c_n-1) \bmod n}}^{n-1} \omega^{-k(\ell+1)} x^{\ell+i+1} y^2 \otimes x^j$$

$$- \frac{1}{3n^2} \sum_{\substack{\ell,i,j=0 \\ i+j\equiv (c_n-1) \bmod n}}^{n-1} \omega^{-k(\ell+1)} (x^{\ell+i+1} y \otimes x^i y + x^{\ell+i} y^2 \otimes x^j y^2)$$

$$= -\frac{1}{3n^2} f_{n,k} \sum_{\substack{\ell,r,s=0 \\ r+s\equiv \ell \bmod n}}^{n-1} \omega^{-k\ell} x^r \otimes x^s + \frac{1}{3n^2} f_{n,k} \sum_{\substack{\ell,r,s=0 \\ r+s+1\equiv \ell \bmod n}}^{n-1} \omega^{-k\ell} x^r y^2 \otimes x^s$$

$$+ \frac{1}{3n^2} f_{n,k} \sum_{\substack{\ell,r,s=0 \\ r+s+1\equiv \ell \bmod n}}^{n-1} \omega^{-k\ell} x^r \otimes x^s y^2 - \frac{1}{3n^2} f_{n,k} \sum_{\substack{\ell,r,s=0 \\ r+s+2\equiv \ell \bmod n}}^{n-1} \omega^{-k\ell} x^r y^2 \otimes x^s y^2,$$

and so the result follows.

Finally, **(c)** follows straightforwardly. For example,

$$\begin{aligned} m_{n,1}(E_{k,0}) &= m_{n,1}(p_k) - \omega^{-k} m_{n,1}(p_k y^2) \\ &= u p_k x^{n/2} + w p_k x^{(n-2)/2} y^2 - \omega^{-k} u p_k x^{n/2} y^2 - \omega^{-k} w p_k x^{(n-2)/2+1} y^2 \\ &= u p_k x^{n/2} - \omega^{-k} u p_k x^{n/2} y^2 \\ &= u\omega^{kn/2}(p_k - \omega^{-k} p_k y^2) = u\omega^{kn/2} E_{k,0}, \end{aligned}$$

for n even, and the rest are similar. □

Theorem. *(See [CT26a, Theorem 7B.9]). For the complete graph K_4,*

$$\begin{aligned} Mh_{n,1}(K_4) = n\big(n^3(40b+41) - 6n^2(13b+14) + 11n(4b+5) - 6(b+2)\big)t^4 \\ + n\big(n^2(13b+14) - 3n(4b+5) + 2(b+2)\big)t^2, \end{aligned}$$

and for the complete bipartite graph $K_{3,2}$,

$$Mh_{n,1}(K_{3,2}) = n\big(n^2(13b+14) - 3n(4b+5) + 2(b+2)\big)(t^6 + 3t^4 + 3t^2 + 1).$$

Proof. We begin by inspecting the circle counts associated with the hypercube of states of the blowup of K_4, obtained using the code described in [CT26a, CT26b]:

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circle counts by homological degree: [[2], [1, 2, 1, 1, 2, 1], [1, 2, 2, 1, 2, 1, 1,
    3, 1, 2, 1, 2, 1, 2, 1], [2, 1, 2, 1, 3, 1, 3, 2, 3, 1, 1, 2, 1, 2, 2, 2, 1, 3,
    2, 1], [2, 1, 2, 1, 2, 2, 2, 4, 2, 2, 2, 2, 1, 1, 2], [1, 3, 1, 1, 3, 1], [2]]
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The maximum number of circles of any state is 4, and so by using [CT26a, Remark 7B.2], we only need to solve for the coefficients of a degree four polynomial in n without a constant term,

$$c_1 n^4 + c_2 n^3 + c_3 n^2 + c_4 n,$$

for each degree of t up to six and b up to one, using the data from Table 1 for K_4 up to $n = 4$. The only nontrivial systems of equations that require solving are

$$\begin{aligned} \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 &= 0 \\ 16\alpha_1 + 8\alpha_2 + 4\alpha_3 + 2\alpha_4 &= 180 \\ 81\alpha_1 + 27\alpha_2 + 9\alpha_3 + 3\alpha_4 &= 1512 \\ 256\alpha_1 + 64\alpha_2 + 16\alpha_3 + 4\alpha_4 &= 5928 \end{aligned} \qquad \begin{aligned} \beta_1 + \beta_2 + \beta_3 + \beta_4 &= 0 \\ 16\beta_1 + 8\beta_2 + 4\beta_3 + 2\beta_4 &= 180 \\ 81\beta_1 + 27\beta_2 + 9\beta_3 + 3\beta_4 &= 1512 \\ 256\beta_1 + 64\beta_2 + 16\beta_3 + 4\beta_4 &= 5952 \end{aligned}$$

4	0	$4(15b+18)(3t^2+1)$	$24(247b+248)t^4+24(27b+28)t^2$
3	0	$3(11b+13)(3t^2+1)$	$504(3b+3)t^4+3(83b+85)t^2$
2	0	$2(7b+8)(3t^2+1)$	$20(3b+3)(3t^4+t^2)$
1	0	$(3b+3)(3t^2+1)$	$(3b+3)t^2$
n / m	2	3	4

TABLE 1. $Mh_{n,1}(K_m)$ for a range of m, n.

4	0	0	$4(15b+18)(t^4+6t^2+1)$	$24(27b+28)(t^6+3t^4+3t^2+1)$
3	0	0	$3(11b+13)(t^4+6t^2+1)$	$3(83b+85)(t^6+3t^4+3t^2+1)$
2	0	0	$2(7b+8)(t^4+6t^2+1)$	$20(3b+3)(t^6+3t^4+3t^2+1)$
1	0	0	$(3b+3)(t^4+6t^2+1)$	$(3b+3)(t^6+3t^4+3t^2+1)$
n / (m,p)	$(1,1)$	$(2,1)$	$(2,2)$	$(3,2)$

TABLE 2. $Mh_{n,1}(K_{m,p})$ for a range of m, p, n.

associated with degree one and zero in b respectively for degree four in t, and similarly for degree two in t:

$$\begin{aligned} \delta_1+\delta_2+\delta_3+\delta_4 &= 3 & \gamma_1+\gamma_2+\gamma_3+\gamma_4 &= 3 \\ 16\delta_1+8\delta_2+4\delta_3+2\delta_4 &= 60 & 16\gamma_1+8\gamma_2+4\gamma_3+2\gamma_4 &= 60 \\ 81\delta_1+27\delta_2+9\delta_3+3\delta_4 &= 249 & 81\gamma_1+27\gamma_2+9\gamma_3+3\gamma_4 &= 255 \\ 256\delta_1+64\delta_2+16\delta_3+4\delta_4 &= 648 & 256\gamma_1+64\gamma_2+16\gamma_3+4\gamma_4 &= 672 \end{aligned}$$

The reader is encouraged to check that the solutions are given by

$$(1) \qquad \begin{pmatrix}\alpha_1\\ \alpha_2\\ \alpha_3\\ \alpha_4\end{pmatrix} = \begin{pmatrix}40\\ -78\\ 44\\ -6\end{pmatrix}, \quad \begin{pmatrix}\beta_1\\ \beta_2\\ \beta_3\\ \beta_4\end{pmatrix} = \begin{pmatrix}41\\ -84\\ 55\\ -12\end{pmatrix}, \quad \begin{pmatrix}\delta_1\\ \delta_2\\ \delta_3\\ \delta_4\end{pmatrix} = \begin{pmatrix}0\\ 13\\ -12\\ 2\end{pmatrix}, \quad \begin{pmatrix}\gamma_1\\ \gamma_2\\ \gamma_3\\ \gamma_4\end{pmatrix} = \begin{pmatrix}0\\ 14\\ -15\\ 4\end{pmatrix}.$$

For $K_{3,2}$, the circle counts associated with the hypercube of states of the blowup are given by:

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circle counts by homological degree: [[3], [2, 2, 2, 2, 2, 2], [3, 1, 1, 1, 1, 1, 1,
    1, 1, 3, 1, 1, 1, 1, 3], [2, 2, 2, 2, 2, 1, 1, 1, 1, 2, 2, 1, 1, 1, 1, 2, 2, 2,
    2, 2], [3, 1, 1, 1, 1, 3, 1, 1, 1, 1, 1, 1, 1, 1, 3], [2, 2, 2, 2, 2, 2], [3]]
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As above, we only need to solve for the coefficients of a degree three polynomial in n without a constant term, i.e. of the form $c_1n^3+c_2n^2+c_3n$, for each degree of t up to six and b up to one, using the data from Table 2 for $K_{3,2}$ up to $n=3$. Comparing with the data for K_4, it is straightforward to see that in fact the solutions for the coefficients δ_i, γ_i, $i=1,\dots,4$ from Equation 1 can be reused to solve the nontrivial systems of equations for $K_{3,2}$, up to overall constant factors. □

REFERENCES

[CT26a] D. W. Collison and D. Tubbenhauer. Categorifying the Penrose polynomial using nonorientable surfaces. 2026. https://arxiv.org/abs/2607.01632.

[CT26b] D. W. Collison and D. Tubbenhauer. Code, data and more for "Categorifying the Penrose polynomial using nonorientable surfaces". 2026. https://github.com/dtubbenhauer/graphhomology.

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